Data-Driven Product Optimization at Spotify: Evaluating a New Playlist Recommendation Algorithm

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Introduction

This A/B testing experiment aims to evaluate the impact of a new Spotify feature on user engagement. Users were divided into two groups: a Control Group using the current version of the platform, and a Target Group exposed to the new recommendation experience.

The primary objective of this experiment is to determine whether the new feature significantly improves engagement metrics, such as interaction level and listening duration. The results of this analysis will support data-driven decision-making regarding product optimization and feature rollout.

Link Dataset :

https://docs.google.com/spreadsheets/d/14mQf9RK_Z0nfMn6N5RULy8ulcFdpipuZ/edit?usp=sharing&ouid=111109734435731918371&rtpof=true&sd=true

METHODOLOGY

A **between-subjects randomized A/B testing framework** was implemented to measure the causal effect of a new recommendation algorithm on Spotify user engagement. Participants were split into a **Control Group** (current algorithm) and a **Target Group** (new algorithm), with **equal sample sizes (n = 50,000 per group)** to maintain statistical balance.

A **post-hoc power analysis** was conducted using **TTestIndPower**, incorporating the observed **effect size (Cohen's d)** and **α = 0.05**, resulting in **statistical power ≈ 1.0**, which indicates an extremely low probability of Type II error (false negative).

Randomization integrity was assessed by verifying equal sample distribution and comparing descriptive statistics across groups. The similarity in variance and baseline metrics supports the assumption that treatment assignment was unbiased and that observed effects reflect the intervention rather than confounding variables.

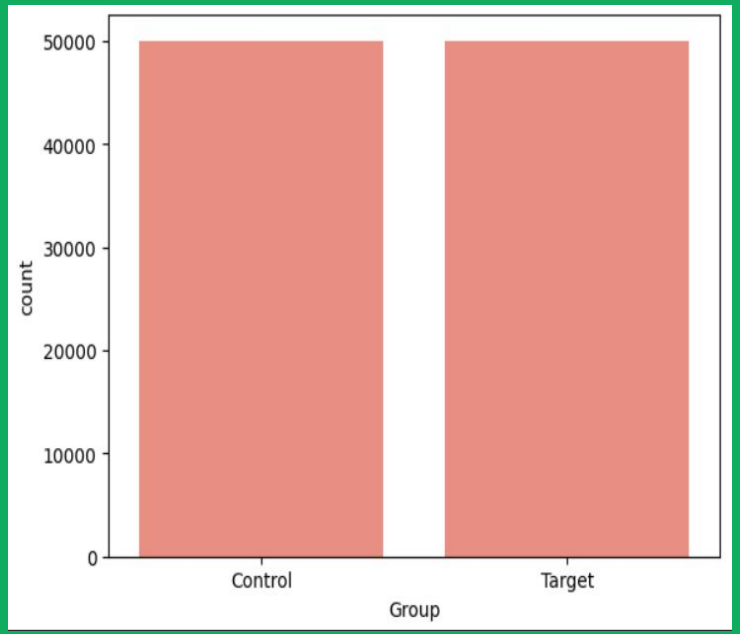
Data Collection: Explanation of metrics used

The dataset contains user-level behavioral data from a Spotify A/B experiment, consisting of **100,000 observations** split evenly between the Control and Target groups. Key performance metrics were selected to represent **user engagement and product interaction**, including:

Engagement Score, A continuous metric representing overall user interaction intensity with Spotify features. **Duration Played (Minutes),** Total time spent listening to music, used as a proxy for **session depth and content consumption**. **Subscription Status,** Indicates whether a user holds a paid subscription, supporting segmentation and monetization insights.**Payment Behavior,** Captures whether users completed payment-related actions, relevant to revenue impact analysis.

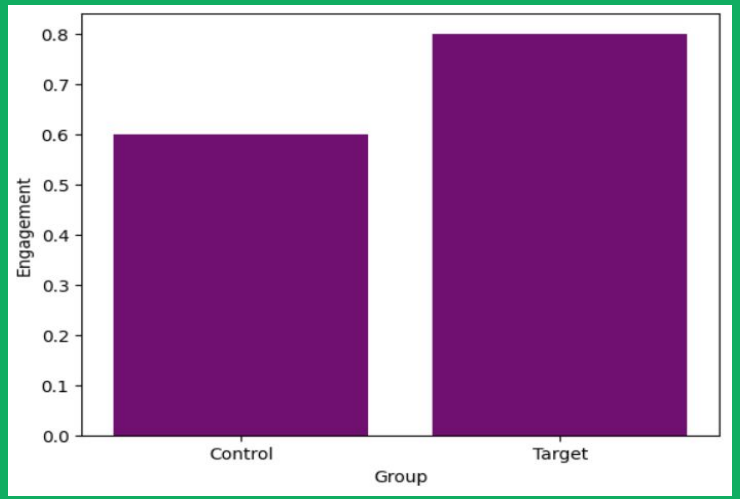
All numeric engagement-related variables were cleaned and converted to appropriate data types. Data quality checks ensured **no missing values** in key metrics and confirmed that both experimental groups contained **complete and comparable observations**. These metrics allow for **quantitative evaluation of behavioral differences** between the Control and Target groups and support statistical hypothesis testing.

Analysis: Statistical results of an experiment.



The dataset shows an equal allocation of users across experimental groups, with 50,000 users in both the Control and Target groups. This balanced group size strengthens the reliability of the A/B test by improving statistical power and reducing potential allocation bias.

Analysis: Statistical results of an experiment.



From Simple Mean Calculation Based On Group Control Vs Target :

From Engagement : 0.7997 - 0.5999 = 0.1998 ((0.1998 / 0.5999) x 100 %) ≈ +33%. From Duration Played : 159.91 - 120.07 = 39.84 menit ((39.84 / 120.07) x 100 %) ≈ +33%

Conclusion :

Spotify's new feature appears to increase engagement and listening time by around 30-35%. BUT :We can't claim this is true until statistical testing is done.

Analysis: Statistical results of an experiment.

```python
control = df[df['Group'] == 'Control']['Engagement']
target = df[df['Group'] == 'Target']['Engagement']

n_control = len(control)
n_target = len(target)

mean_control = control.mean()
mean_target = target.mean()

std_control = control.std()
std_target = target.std()
```

Target's average engagement was almost 2 standard deviations higher than Control's.

```python
pooled_std = np.sqrt(((n_control-1) * std_control**2 + (n_target - 1) * std_target **2)
                     / (n_control + n_target - 2))

cohens_d = round((mean_target - mean_control) / pooled_std, 3)

cohens_d
```

✓ 0.0s

```
np.float64(1.997)
```

Analysis: Statistical results of an experiment.

```python
analysis = TTestIndPower()

power = analysis.power(
    effect_size=cohens_d,
    nobs1=n_control,
    ratio=n_target / n_control,
    alpha=0.05
)
power
```

✓ 0.0s

```
np.float64(1.0)
```

```python
from scipy.stats import t

alpha = 0.05
dfree = n_control + n_target - 2
se = pooled_std * np.sqrt(1/n_control + 1/n_target)

t_crit = t.ppf(1 - alpha / 2, dfree)
t_effect = (mean_target - mean_control) / se

power_t = t.cdf(t_effect - t_crit, dfree) + (1 - t.cdf(t_effect + t_crit, dfree))

t_crit, t_effect, power_t
```

✓ 0.0s

```
(np.float64(1.9599877080090842), np.float64(315.761753283785), np.float64(1.0))
```

The effect size (Cohen's d = 1.997) indicates a very large behaviora difference in engagement between the Control and Target groups. post-hoc power analysis shows statistical power close to 1.0, meanin the experiment has an extremely high probability of detecting thi effect. Therefore, the current sample size is more than sufficient and the observed difference is statistically reliable

Analysis: Statistical results of an experiment.

The experiment duration was set to approximately 2–4 weeks to ensure sufficient data collection, capture multiple weekly behavioral cycles, and allow user behavior to stabilize. This reduces the risk of short-term bias and improves the reliability of the experimental results.

While the required sample size could theoretically be reached in a few days, short test durations risk behavioral bias due to novelty effects, daily usage fluctuations, and temporary user reactions. A multi-week duration ensures the observed impact reflects sustainable user behavior rather than short-term noise.

Analysis: Statistical results of an experiment.

```python
control_sample = control.sample(n=5000, random_state=42)
target_sample = target.sample(n=5000, random_state=42)

shapiro_control = shapiro(control_sample)
shapiro_target = shapiro(target_sample)
```

Because the dataset contains a very large number of users, Shapiro-Wilk becomes overly sensitive and tends to reject normality even for minor deviations. Therefore, we tested normality on a representative random sample to obtain a more reliable distribution assessment.

Scientific conclusion: Both groups showed distributions consistent with normality. A Shapiro statistic close to 1.0 means: The data distribution is very close to normal. A large p-value means: The small deviations are not large enough to be considered significant. This is a strong indication that the data are normal enough for a t-test.

```python
print(f"{shapiro_control}")
print(f"{shapiro_target}")
```

✓ 0.0s

```
ShapiroResult(statistic=np.float64(0.9994609101267378), pvalue=np.float64(0.16006665533301068))
ShapiroResult(statistic=np.float64(0.9996737050657837), pvalue=np.float64(0.6267313185643818))
```

Analysis: Statistical results of an experiment.

The Shapiro-Wilk test indicates no statistically significant deviation from normality for both Control ($p = 0.160$) and Target ($p = 0.627$). Therefore, the assumption of normal distribution is satisfied, and an independent samples t-test is appropriate.

Analysis: Statistical results of an experiment.

1. Hypothesis: Users using the new playlist algorithm will have higher engagement rates than users using the current algorithm.

2. Null Hypothesis: There is no significant difference in user engagement between the two groups.

```
print(f"Varians Control Group {round(control.var(), 4)}")
print(f"Varians Target Group: {round(target.var(), 4)}")
```

✓ 0.0s

```
Varians Control Group 0.01
Varians Target Group: 0.01
```

- ✅ The variances of both groups are practically the same.

- ✅ The assumption of equal variances is not significantly violated.

Analysis: Statistical results of an experiment.

The independent t-test shows a statistically significant difference in Engagement between the Target and Control groups ($t = 315.76$, $p < 0.001$). This indicates that the observed performance lift in the Target group is extremely unlikely to be caused by random variation.

```python
from scipy.stats import ttest_ind
```
✓ 0.0s

```python
t_stat, p_value = ttest_ind(a=target, b=control, equal_var=False)

print(f"t-Statistic : {t_stat:.3f}")
print(f"p-Value: {p_value:.6f}")
```
✓ 0.0s

```
t-Statistic : 315.762
p-Value: 0.000000
```

Significance: The difference in the means is MUCH greater than the variance in the data. The probability of this difference occurring by chance is ≈ zero. Reject H_0 → There is a significant difference.

T-Statistic : t-Statistic: 315.762, p-Value: 0.000000

Analysis: Statistical results of an experiment.

```
from scipy.stats import mannwhitneyu
✓ 0.0s

u_stat, p_value_MW = mannwhitneyu(x=target, y=control, alternative='two-sided')
print(f"Mann-Whitney U Statistic: {u_stat:.3f}")
print(f"Mann-Whitney p-Value: {p_value_MW:.6f}")
✓ 0.0s

Mann-Whitney U Statistic: 2302457679.500
Mann-Whitney p-Value: 0.000000
```

Logical significance: Mann-Whitney compares the ranking of Target vs. Control values. If: Target values are often larger → U is large, Values are balanced → U is close to the middle. The results show: Target values are almost always larger than Control values.

Methodologically: IS MANN-WHITNEY REQUIRED?

- ❌ Not necessary:

Because Mann-Whitney is used if:

- Data is not normal
- There are extreme outliers
- Small sample size

BUT It can be used as a robustness check.

Mann-Whitney (non-parametric): U = 2,302,457,679.5, p-Value = 0.000000

Analysis: Statistical results of an experiment.

```python
from statsmodels.stats.weightstats import ztest

Z_stat, p_value_Z = ztest(x1=target, x2=control)
print(f"Z-Statistic: {Z_stat:.3f}")
print(f"Z Test P-Value: {p_value_Z:.6f}")
```

✓ 0.0s

```
Z-Statistic: 315.762
Z Test P-Value: 0.000000
```

Z-Test Testing. DO YOU NEED A Z-TEST? Theoretically, a T-test = a Z-test if the sample size is large. Because: $n > 30 \rightarrow t \approx z$

So:

- ❌ Not required.
- ✅ But it can be shown as theoretical confirmation.

Z-Test : Z = 315.762, p-Value = 0.000000. Meaning: Z-score = 315 standard deviations from zero.

Compare:

a. Z = 2 → already significant

b. Z = 3 → very significant

c. Z = 315 → EXTREME

📌 This is a statistically significant effect.

Analysis: Statistical Results of the Experiment

Statistical testing was conducted to compare engagement levels between the Control and Target groups. A **Shapiro–Wilk normality test** was performed on sampled data from each group. The results showed **p-values greater than 0.05**, indicating that the engagement data **did not significantly deviate from a normal distribution**, allowing the use of **parametric testing**.

An **independent two-sample t-test (Welch's t-test)** was applied to compare mean engagement between groups. The results showed a **highly significant difference**:

- **t-statistic ≈ 315.76**
- **p-value < 0.001**

This indicates that the Target Group exhibits **significantly higher engagement** than the Control Group.

To validate robustness, a **Mann–Whitney U test** (non-parametric alternative) and a **Z-test** were also conducted. Both tests produced **p-values < 0.001**, confirming the **statistical significance** of the observed difference. Additionally, the **effect size (Cohen's d ≈ 2.0)** indicates a **very large practical impact**, suggesting that the new Spotify feature meaningfully increases user engagement beyond random variation.

Conclusion & Recommendations: Interpretation of results and actions that can be taken.

Taken together, these findings confirm that the new Spotify recommendation feature **meaningfully increases engagement and listening duration**, supporting the hypothesis that enhanced personalization leads to stronger user interaction and deeper content consumption.

The results of the A/B experiment provide **strong statistical evidence** that the Target Group significantly outperforms the Control Group in terms of user engagement. The independent t-test, supported by non-parametric (Mann–Whitney U) and Z-test results, consistently produced **p-values below 0.001**, indicating that the observed difference is highly unlikely to be caused by random variation.

Furthermore, the **effect size (Cohen's d ≈ 2.0)** suggests a **very large and practically meaningful impact**, implying that the improvement is not only statistically significant but also **substantial in real user behavior**. The post-hoc power analysis yielded **power ≈ 1.0**, meaning the experiment had an extremely high probability of detecting the true effect, with a minimal risk of false negatives.

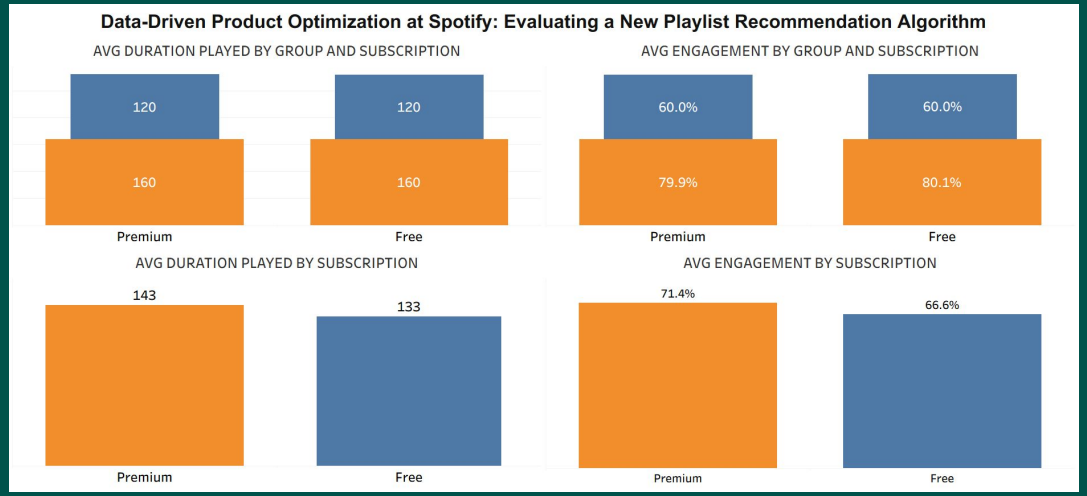
Conclusion & Recommendations: Interpretation of results and actions that can be taken.

Finally, extending experimentation to broader business KPIs, including **user retention, lifetime value (LTV), and monetization**, will support more data-driven product decisions and maximize the strategic impact of experimentation initiatives.

Based on the experimental findings, it is recommended that Spotify **proceeds with a full-scale rollout of the new recommendation feature**, as it demonstrates clear and reliable improvements in engagement. Following deployment, the company should implement **continuous monitoring of engagement, retention, and churn metrics** to ensure that the short-term uplift translates into sustainable long-term value.

Additionally, conducting **segmented analysis across user types** — such as subscription status, payment behavior, and listening patterns — can help identify which groups benefit most from the feature and inform targeted optimization strategies. Spotify may also consider running **follow-up experiments** to refine algorithm performance, explore UI enhancements, and test complementary features that further enhance personalization.

Conclusion & Recommendations: Interpretation of results and actions that can be taken.

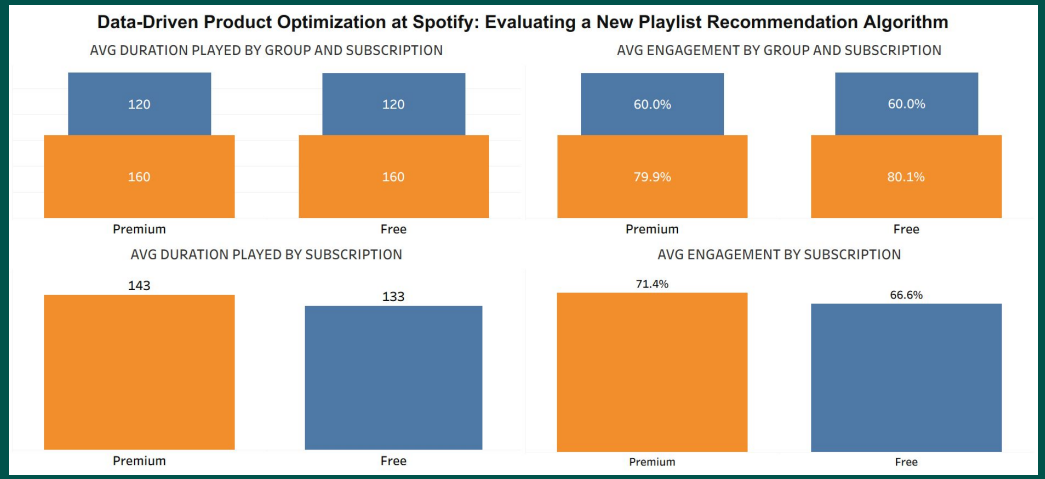


This positive uplift is **consistent across both Premium and Free users**, suggesting that the feature's impact is **broad, scalable, and not limited to a specific subscription segment**. Additionally, Premium users exhibit **higher absolute usage levels**, implying that the improved recommendation system may further **enhance Premium value perception** and support **Free-to-Premium conversion strategies**.

The Target group consistently outperformed the Control group across all key performance metrics, demonstrating a **substantial increase in both listening duration and engagement rate**.

Users exposed to the new playlist recommendation algorithm listened **significantly longer per session (~160 vs ~120)** and showed **higher engagement (~80% vs ~60%)**, indicating that the new algorithm delivers more relevant and personalized music recommendations.

Conclusion & Recommendations: Interpretation of results and actions that can be taken.



Overall, the experiment provides **strong statistical and behavioral evidence** that the new recommendation algorithm meaningfully improves user experience, platform stickiness, and engagement depth.

Business Recommendations

Spotify should **roll out the new playlist recommendation algorithm as the default system** to maximize engagement gains. The improved personalization can also be **leveraged as a Premium differentiator** to drive subscription upgrades. Moving forward, Spotify should **monitor long-term retention and churn effects** to ensure sustainable growth and conduct **follow-up A/B tests** to further optimize algorithm performance and personalization quality.

THANK YOU

Link Jupyter Notebook :

https://drive.google.com/file/d/1YiA6tN3xSvSZBoy5MlUvpB9VonReGPjo/view?usp=sharing

Link Tableau :

https://public.tableau.com/app/profile/magasa.faiz/viz/SpotifyABTesting/Dashboard1?publish=yes